

Survey Imputation: MICE vs KNN

Comparing traditional missing data imputation methods on a 2024 U.S. presidential election survey dataset with MNAR (Missing Not At Random) missingness patterns.

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Authors: Yuchen Ding, Angelina Lu, Weiyushi Tian, Brady West

What this project does

Survey datasets frequently contain missing responses. This project evaluates two traditional imputation approaches under MNAR conditions — a realistic and challenging missingness mechanism where the probability of a value being missing depends on the value itself.

Methods compared:

Method	Implementation	Key parameter
MICE (PMM)	<code>mice</code> R package	m=5 imputations, maxit=5
KNN	<code>VIM::kNN</code>	k selected via accuracy grid search

Data conditions:

- `self_mnar` — self-reported missingness pattern
 - `cross_mnar` — cross-item missingness pattern
 - `cleaned` — complete-case baseline
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Key finding

KNN with optimally selected k achieved **95–96% imputation accuracy** on held-out missing cells across three survey question blocks. MICE (PMM) produced comparable distributional fidelity. Trade-offs between the two methods are context-dependent — see the full paper for details.

Repository structure

survey-imputation/

```
|— R/
| |— 01_mice_imputation.Rmd # MICE pipeline + visualization
| |— 02_knn_imputation.Rmd # KNN with optimal k selection
|— data/
| |— generate_demo_data.R # Synthetic data generator (run this first)
|— README.md
```

Data note: Original survey data are not included (restricted access). Run [data/generate_demo_data.R](#) to generate a compatible synthetic dataset.

How to run

Step 1: generate synthetic data

```
source("data/generate_demo_data.R")
```

Step 2: open and knit either notebook in RStudio

R/01_mice_imputation.Rmd

R/02_knn_imputation.Rmd

Dependencies: [mice](#), [VIM](#), [caret](#), [ggplot2](#), [dplyr](#)

```
install.packages(c("mice", "VIM", "caret", "ggplot2", "dplyr"))
```

Related work

This analysis is part of a broader project comparing LLM-based and traditional imputation methods. The LLM comparison component is described in the AAPOR 2025 paper.